

OpenClaw Plant Care Assistant Tutorial

Before starting, please complete the [01-Hardware Configuration], [02-openclaw API-KEY Configuration], [03-openclaw Model Switching], [04-AI Voice Interaction Configuration], [05-Three Dialogue Solution Configuration Tests] under the `Prerequisites` directory. If you are currently only using TUI / whatsapp, you can skip `04` for now. This article assumes the basic environment has been prepared and only expands on the plant care project itself.

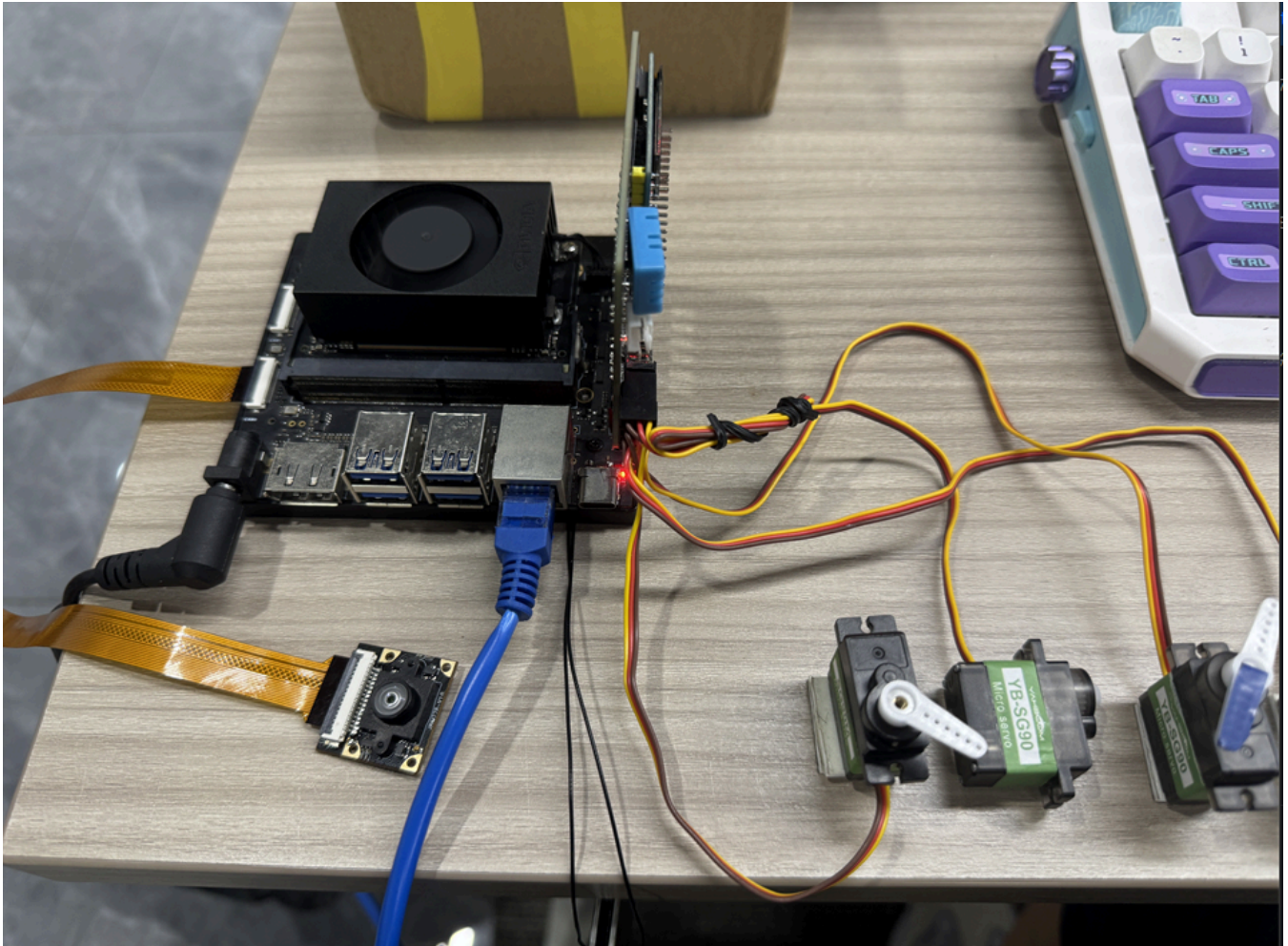
1. Tutorial Objectives

This article demonstrates OpenClaw's multi-peripheral coordination capability through the plant care scenario.

- Understand how to combine temperature/humidity, OLED, RGB, and camera into a plant care assistant
- Communicate with OpenClaw through three methods: whatsapp, TUI, and voice code

2. Hardware Preparation

- OpenClaw mainboard
- DHT temperature and humidity sensor
- OLED display
- RGB light strip / RGB light bead module
- CSI camera
- `[Optional] AI voice module + speaker`
- Wiring diagram



3. Software Preparation

Code entry points that will be used in the tutorial

- Camera photo command: `/home/jetson/hardware_ha1_demo/seewhat`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`

This project is essentially not dependent on a single "large project script", but lets OpenClaw orchestrate multiple basic tools:

- `dht` is responsible for reading environmental temperature and humidity
- `oled` is responsible for displaying care information
- `rgb` is responsible for status alerts
- `seewhat` is responsible for observing plant appearance

4. Test Project Basic Capabilities

Before starting this project, please complete one `openclaw tui` basic dialogue self-check; if you haven't done it yet, see [05-Three Dialogue Solution Configuration Tests]. After passing, enter TUI to verify the basic capabilities in the project.

Terminal input:

```
openclaw tui
```

```

openclaw tui - ws://127.0.0.1:18789 - agent main - session main

session agent:main:main
connected | idle
agent main | session main (openclaw-tui) | baillian/qwen3.5-plus | tokens 0/1.0m (0%)

```

After confirming OpenClaw can dialogue normally, you can try the following types of commands:

- Read current temperature and humidity
- Display "Plant OK" on OLED
- Change RGB light to green
- Help me take a photo

If these individual capabilities can all run through, it indicates that the main chains required by the project are all prepared.

5. Three Solutions for Communicating with OpenClaw

Solution	Suitable Scenarios	Advantages	Recommended Positioning
whatsapp	Remote care, classroom demonstrations	Easy to connect, suitable for remote status viewing	Demonstrate "remote natural language care"
TUI	Local debugging, project joint debugging	Most direct, easiest to troubleshoot	First run through multi-peripheral chain
Voice code	Voice companionship, complete AI experience	Closer to real assistant experience	As final demonstration solution

If optional AI voice module + speaker are selected, whatsapp dialogue, tui, whatsapp dialogue need to broadcast OpenClaw's reply content. You can run the following program in terminal. AI voice module dialogue does not need to run repeatedly

- Modify whether to enable broadcast parameter

```

/home/jetson/voice_to_openclaw/.env

```

```

ENABLE_TTS=true #This parameter controls whether to broadcast. After modification and saving, run the following program in terminal

```

- Terminal input

```

cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts

```

```
(.venv) jetson@yahboom:~/voice_to_openclaw$ python -m src.openclaw_session_tts  
[系统] OpenClaw 会话播报器已启动  
[系统] 监听目录=/home/jetson/.openclaw/agents/main/sessions  
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 whatsapp Solution

For whatsapp connection and robot creation steps, please directly refer to [05-Three Dialogue Solution Configuration Tests]. This section only retains dialogue examples suitable for the plant care project.

After successfully creating a robot, you can remotely inquire about plant status in whatsapp, for example:

- Take a photo to see the current plant status, and perform plant care

20:10



< 2

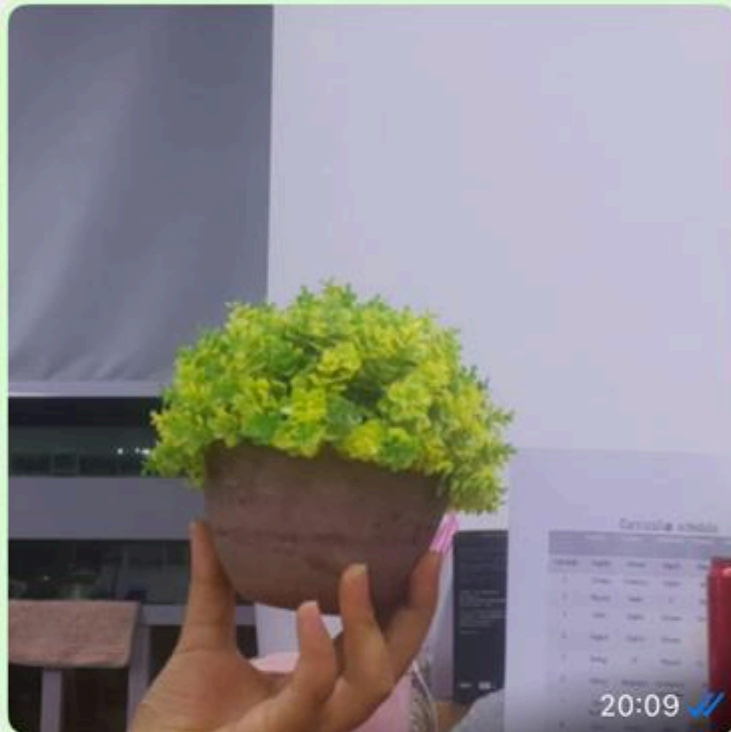


+86 153 3885 7526 (你)

给自己发消息

Take a photo to see the current plant status and perform plant care

20:09 ✓✓



20:09 ✓✓

I can see the plant! It's a small potted plant with vibrant green-yellow leaves. The foliage looks full and healthy. Let me complete the plant care workflow.

20:10 ✓✓

Plant care check completed!

Status:

- 📷 **Visual check:** Plant looks healthy with vibrant green-yellow foliage
- 🌡️ **Temperature:** 29.7°C (85.5°F)
- 💧 **Humidity:** 52%
- 📺 **OLED:** Displaved "Plant:"





5.2 TUI Solution

If you have completed the TUI self-check in Chapter 0, directly enter terminal text dialogue:

```
openclaw tui
```

Then you can directly enter project-related commands, for example:

- Plant care

A screenshot of a TUI application. The top half shows a video feed of a lush green artificial pine tree. Below the video is a terminal window with a dark background and light green text. The terminal has tabs for 'Issues', 'Output', 'Debug Console', 'Terminal', and 'Port 1'. The terminal output shows environmental data from a DHT11 sensor, a command prompt '[Plant Care]', and a detailed report on the plant's status, including temperature, humidity, and care instructions. At the bottom, it shows the session information: 'agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 2/1.0m'.

5.3 Voice Code Solution

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up process, please directly refer to [04-AI Voice Interaction Configuration]. After waking up, you can directly say control content, for example:

- Help me check the plant status and whether it's suitable for watering now

```
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音, 正在转写

[你] Me check the plant status and whether it's suitable for watering now.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Let me analyze the plant's condition and check the environment to
           determine if it needs watering.

## 📋 Plant Status Check
           **Environmental Conditions:**
- **Temperature:** 30.1°C (86.2°F) -
           Quite warm! 🌡️
- **Humidity:** 52% - Moderate

**Plant Analysis:**
           Looking at the captured image, I notice:
- The plant has **bright
           green/yellow foliage** with uniform coloring
- Leaves appear
           **vibrant and upright** (no wilting visible)
- The foliage has an
           **artificial appearance** (plastic-like sheen)

**🚰 Watering
           Recommendation:**
```

6. Project Implementation

Note: The following are all dialogue demonstrations using `openc1aw tui` method. You can also choose whatsapp or voice method to complete.

6.1 Scenario One: Environment Reading

Experimental Objective

Read the current temperature and humidity around the plant, completing the most basic environmental perception.

Effect Description

OpenClaw will call DHT to read current environmental data, and use natural language to feedback temperature and humidity.

Example Instructions

- Read temperature and humidity near the current plant
- What is the current environmental temperature and humidity
- Help me check if the potted plant area is too dry

6.2 Scenario Two: Status Judgment

Experimental Objective

Let OpenClaw provide a brief plant care suggestion after obtaining environmental data.

Effect Description

Besides directly reporting values, OpenClaw can also summarize conclusions like "too dry", "too hot", "status normal" in more natural everyday language.

Example Instructions

- Judge plant status based on current temperature and humidity
- Is it suitable for watering now

6.3 Scenario Three: Audio-Visual Alert

Experimental Objective

When environmental status is not ideal, use OLED and RGB to give more intuitive alerts.

Effect Description

For example, when environment is normal, light green; when too hot, light yellow; when abnormal, light red. At the same time, display simple prompt information on OLED, so it's not just "able to speak", but has clear peripheral feedback.

Example Instructions

- If environment is normal, light green
- If too hot, light yellow and prompt me
- Display "Plant Needs Care" on OLED

Extensible Gameplay

- Link with buzzer for stronger alerts
- Make into desktop care status light
- Add scheduled polling and auto-refresh

6.4 Scenario Four: Photo Observation

Experimental Objective

Use camera to photograph current plant image, and let OpenClaw make a simple description.

Effect Description

This step is suitable for observing leaves, pot surface, or overall growth. The current tutorial focuses on "taking a look and making a description", not automatic disease judgment, so it is more suitable for teaching demonstrations and lightweight auxiliary observation.

Example Instructions

- Take a photo of the plant
- Check how the leaves are doing
- After taking a photo, describe the current plant image